

Modeling and simulation of turbulent nuclear flames in Type Ia supernovae

Arash G. Nouri^a, Peyman Givi^a, Daniel Livescu^{b,*}

^a Department of Mechanical Engineering and Materials Science, University of Pittsburgh, Pittsburgh, PA, 15261, USA

^b Los Alamos National Laboratory, Los Alamos, NM, 87544, USA

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ABSTRACT

A review is conducted of the state of progress in physical modeling and computational simulations of turbulent flames as pertaining to Type Ia supernovae. The fundamental governing equations are provided, along with various assumptions made for their simplifications; and the numerical means of their discretizations. A case is made for a more systematic inclusion of turbulence, including turbulence-reaction interactions, for future investigations.

1. Introduction

Theoretical modeling and computational simulations of Type Ia supernovae (SNe Ia) have been the subject of widespread investigations for almost 60 years now [1–5]. The subject has experienced even more recent interest due to the utilization of SNe Ia as a distance indicator to determine the local expansion rate, and to map the expansion history of the Universe [6–11]. These investigations have led to the discovery of the accelerated expansion [8,12], confirming the connections between cosmological redshifts and time dilation [13–15], and are providing a better understanding of the constraints for the dark energy equation of state parameter [16–18]. The literature is rich with comprehensive reviews on SNe Ia including their theoretical description [19–22], spectral properties [23], utilization for measuring the Hubble constant [7], and photometry in infrared (IR) and optical bands [24,25], amongst other applications.

It is now widely accepted that SNe Ia are most likely the result of a thermonuclear explosion happening in carbon–oxygen (C+O) white dwarfs (WDs) [26–29]. Typical SNe Ia also have a clear correlation, named Phillips relationship, between their peak luminosity, the rate of decline after maximum light, and the color at maximum [30]. However, there is substantial evidence suggesting that there is a distinct variation among SNe Ia [30–33]. The observational diversity pertaining to ranges in luminosity, light-curve shape, color, spectral evolution, and host galaxy environment [34] implies that SNe Ia possibly encompass the end products of very different evolutionary paths [35]. Since the progenitor system of a SN Ia should involve an exploding C+O WD, the proposed evolutionary scenarios are: (i) single, isolated WD (Fig. 1 (a)) which explodes as a result of a thermonuclear runaway initiated by pycno-nuclear reactions in the presence of trace

amounts of light elements [36,37], (ii) single-degenerate (SD) (some possibilities are depicted in Fig. 1(b, c, d)), in which the WD grows in mass through accretion from a non-degenerate companion that can be a main-sequence star (MS), a subgiant, a red giant (RG), an asymptotic giant-branch (AGB) star, or a helium (He) star (e.g. Ref. [38]), (iii) core-degenerate (CD) (Fig. 1(e)), in which the WD plunges into the envelope of a giant companion and merges with the giant's core [39,40], (iv) double-degenerate (DD) (Fig. 1(f)), in which two WDs in a binary system merge smoothly or violently [41–45], and (v) triple-system (Fig. 1(f)), in which two WDs are driven to collide together in a triple star system (e.g. Refs. [46–48]). Each of these scenarios have their own strengths and weaknesses in describing the physics of SNe Ia, with those depicted in Fig. 1(b), (c), and (f) considered the most likely scenarios [29]. For more detailed reviews on the progenitors of SNe Ia, please see Refs. [29,35,49].

In each one of the progenitor scenarios, the resultant would be a Chandrasekhar mass (M_{Ch}), a sub-Chandrasekhar mass [49], or a super-Chandrasekhar mass (super- M_{Ch}) WD [50]. The M_{Ch} WD-model has been favored in describing a majority of SNe Ia consisting of C+O WDs [51–55], the sub- M_{Ch} WD-model yields a better means of characterizing SNe Ia in WDs with low-mass C+O core mantled by helium [56–59], and super- M_{Ch} WD-model depicts the over-luminous SNe Ia like SN 2003 fg [60,61]. In the M_{Ch} WD-model, the mass of the WD could reach a critical value, M_{Ch} , either by accumulating mass from a binary giant/main sequence companion (Fig. 1(b)) [62,63], or by WD-WD mergers (Fig. 1(f)) [64–68], among other scenarios [69]. As the WD accumulates sufficient materials, the central density of the WD increases and the heat produced by carbon fusion cannot be neutralized by the cooling effect of neutrino emission. A simmering phase continues the evolution with convective carbon burning, lasting for about a century. This phase

* Corresponding author.

E-mail address: livescu@lanl.gov (D. Livescu).

Nomenclature

$\mathbf{u}$	Velocity vector
$\mathbf{x}$	Position vector
$\Delta_{\mathcal{H}}$	LES filter size
δ_{ij}	Kronecker delta
$\dot{\omega}_k$	Species k source term
$\dot{Q}$	Heat source term
γ	Specific heat ratio
λ	Heat diffusivity
M_{kj}	Species k in reaction j
μ	Molecular viscosity
μ_p	Chemical potential
ν_{kj}	Stoichiometric coefficient of species k in reaction j
ρ	Density
τ_{ij}	Viscous stress tensor
Υ_e	Number of electrons per baryon
κ	Kolmogorov length scale
a	Stephan-Boltzmann constant
A_k	Number of nucleons in isotope k
A_m	Mean number of nucleons per isotope
B_k	Binding energy of isotope k , or the species enthalpy
c	Speed of light
D_k	Diffusion coefficient of species k
E	Energy
e	Internal energy
$f_{k,i}$	Volume force acting on species k in direction i
h	Planck's constant
$J_{j,k}$	j^{th} component of the diffusive mass flux vector of species k
k_B	Boltzmann constant
$K_{f,j}$	Forward rate of reaction j

$K_{r,j}$	Backward rate of reaction j
M	Mach number
m_e	Electron rest mass
m_p	Proton rest mass
M_r	Number of reactions
$M_{\odot}$	Solar mass
N	Number of species
N_A	Avogadro number ($6.02252 \times 10^{26} \#particles/kmol$)
n_k	Number density of isotope k , measured as # of particles of isotope k per unit volume
N_{ele}	Number density of free electrons
N_{pos}	Number density of free positrons
p	Pressure
Q'_j	Rate of progress of reaction j
q_j	Heat flux
s	Entropy
T	Temperature
t	Time
V_i^c	i^{th} component of correction diffusion velocity
$V_{k,i}$	i^{th} component of diffusion velocity of species k
W	Mean molecular weight of the mixture
W_k	Molecular weight of species k
X_k	Molecular fraction of species k
Y_k	Mass fraction of species k
Z_k	Number of protons in isotope k
$\mathcal{F}$	Helmholtz free energy
$\mathcal{H}$	LES filter function
$\mathcal{R}$	Universal gas constant ($=N_A k_B$)
$\mathcal{S}_{ij}$	Viscous stress tensor

ends when the temperature rises to the point where convection can no longer transport the carbon burning energy away. At this stage, a thermonuclear flame is initiated. This flame propagates outward and disrupts the WD [70,71]. Once the thermonuclear flame is ignited, there are two possible modes of propagation: deflagration and detonation. A prompt ignition of the detonation flame is disfavored because it does not produce sufficient intermediate mass elements, such as Mg, Si, S, and Ca, [9,72–74]. Therefore, the explosion is believed to start with a deflagration. This stage may become either strongly turbulent and last until the end of the explosion (the deflagration model) [70,75]; or be followed by a delayed detonation (deflagration to detonation transition-DDT) [3,76–79]. Moreover, as SNe Ia are believed to have been the main producers of iron-peak elements, the abundance ratios of Ni and Fe isotopes like $^{58}\text{Ni}/^{56}\text{Fe}$ and $^{54}\text{Fe}/^{56}\text{Fe}$ in the ejecta should agree with their solar system counterparts. These are difficult constraints to satisfy in many of the detonation and deflagration models [74].

The radioactive decay of ^{56}Ni through ^{56}Co to ^{56}Fe produces gamma-ray photons that are absorbed and dominate the heating and thus the luminosity of the ejecta at intermediate times (several weeks) to late times (several months) [80]. It has been shown that different initial conditions and explosion mechanisms predict an approximate range of 0.3–1.1 solar masses for ^{56}Ni yields [33]. A strong ignition with multiple symmetrically distributed “sparks” reduces the ^{56}Ni produced in the subsequent detonation phase because of large amount of burning in deflagration phase. On the other hand, a weak or asymmetrical ignition with little deflagration burning synthesizes abundant ^{56}Ni in detonation phase [33]. The gamma-ray line fluxes, hard X-ray continuum flux spectrum, flux band ratios, and the ratio of the hard X-ray fluxes to those of the gamma-ray lines are sensitive to (and thus can be used as diagnostic of) the total mass of $^{56}\text{Ni}/^{56}\text{Co}$ ejected, ^{56}Ni spatial distribution, kinetic energy and expansion speed, explosion asymmetries, and the distinction between detonation and deflagration [81].

The aim of mathematical/computational models is to capture the

physics of SN explosion, and to elucidate the various possible scenarios on burning and ejecta. Accurate calculations should also help in explaining the observations, and to enhance the distance estimates. The accuracy of predictions obviously depends on the reliability of the model and the precision of the computational solver. Here, a review is conducted of the state of progress in physical & mathematical modeling, and computational simulations of SNe Ia. The bulk of the modeling work in this arena has been done within the astrophysics community. These models go from the early one-dimensional to recent multi-dimensional descriptors. Parallel to these contributions, there has been significant progress within the aerospace research community in development of models for turbulence and turbulence-reaction kinetics interaction. In this survey, a case is made for a more concentrated effort to take advantage of these developments. In order to provide a framework for subsequent discussion, in §2, the general equations of turbulent reactive flows, as have been used for SNe Ia calculations, are provided. Then, an overview is presented of the various simplifying assumptions typically made in tackling the problem computationally. In §3, a review is provided of the various computational simulation strategies, along with presentation of some examples of the physics captured by such simulations. Section 4 is devoted to large eddy simulation (LES) which is considered to be the optimum means of capturing a broad range of flow scales of SNe Ia. This section contains a review of the sub-grid scale (SGS) closures, with appraisals of their prospects for LES of SNe Ia. The role of direct numerical simulation (DNS) is reviewed in §5, as such simulations have proven very effective in both model developments and in understanding the flow physics. Finally, some suggestions for future work are given in §6.

2. Formulation

The index notation is adopted with the (Einstein's) summation rule over repeated indices. The exception is the index k , which is reserved

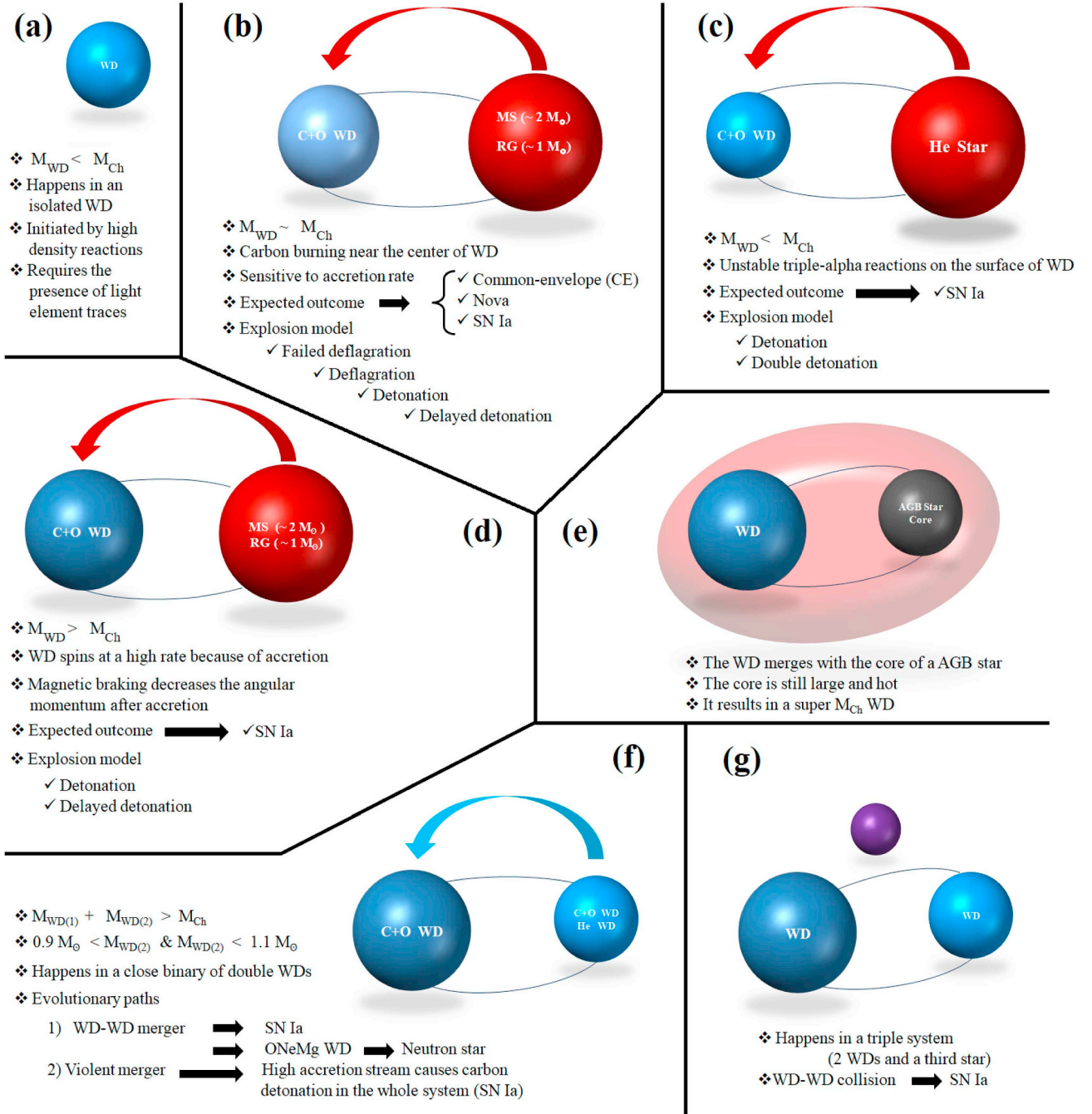


Fig. 1. Evolutionary scenarios for SNe Ia: (a) single, isolated WD, (b) SD M_{Ch} , (c) SD sub- M_{Ch} , (d) SD super- M_{Ch} , (e) CD, (f) DD merger, and (g) triple-system.

for the species, and does not follow the summation rule unless explicitly denoted by the $\sum$ sign.

2.1. Transport

Type Ia supernovae are initiated when the carbon and oxygen ignite in a white dwarf (WD) progenitor, fusing to produce iron-group elements in a thermonuclear runaway. The subsequent deflagration (that may evolve into a detonation) completely disrupts the star. The exploding WD material can be described as a fully ionized plasma with varying degrees of electron degeneracy [82]. The continuum hypothesis is expected to hold for the current scale of the simulations. Therefore, the transport equations of reactive flows should provide a reasonable description of the phenomenon. In the majority of SNe Ia computations to date, only the electron effects on heat conduction are considered, and

the explicit influences of the magnetic field are assumed negligible. This assumption is adopted here, with a recognition of its limitations [83,84].

The set of conservation equations describing the evolution of fully compressible flows with nuclear reactions and thermodynamically active species read:

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho u_j}{\partial x_j} = 0, \quad (1a)$$

$$\frac{\partial \rho u_i}{\partial t} + \frac{\partial \rho u_i u_j}{\partial x_j} = -\frac{\partial}{\partial x_j} [p_{tot} \delta_{ij} - \tau_{ij}] + \sum_{k=1}^N \rho Y_k f_{k,i}, \quad (1b)$$

$$\frac{\partial \rho E_{tot}}{\partial t} + \frac{\partial \rho E_{tot} u_j}{\partial x_j} = -\frac{\partial}{\partial x_j} [u_i (p_{tot} \delta_{ij} - \tau_{ij}) + q_j] + \dot{Q} + \sum_{k=1}^N \rho Y_k f_{k,i} (u_i + V_{k,i}), \quad (1c)$$

$$\frac{\partial \rho Y_k}{\partial t} + \frac{\partial \rho Y_k (u_j + V_{k,j})}{\partial x_j} = \dot{\omega}_k, \quad (1d)$$

where $E_{tot} = e_{tot} + \frac{1}{2} u_i u_i$. Written in this form, without specification of the transport and source terms, the equations are very general and can be derived from many-body quantum mechanics [85]. Generally, the form of the transport terms is still open for matter under SNe Ia conditions. Since the Reynolds number is very large ($\sim 10^{14}$ on length scales comparable to the white dwarf star), turbulent transport is expected to dominate over viscous and diffusive effects [82]. As a first order approximation, assuming a Newtonian fluid, with Fick's law of diffusion and Fourier's law of heat conduction, the transport terms are [86]:

$$\tau_{ij} = 2\mu \left(\mathcal{S}_{ij} - \frac{1}{3} \delta_{ij} \mathcal{S}_{ll} \right), \quad \mathcal{S}_{ij} = \frac{1}{2} \left(\frac{\partial u_j}{\partial x_i} + \frac{\partial u_i}{\partial x_j} \right), \quad (2a)$$

$$\rho Y_k V_{k,j} \approx J_{j,k} = -\rho \left(D_k \frac{W_k}{W} \frac{\partial X_k}{\partial x_j} - Y_k V_j \right), \quad (2b)$$

$$q_j \approx -\lambda \frac{\partial T}{\partial x_j} + \sum_{k=1}^N J_{j,k} B_k, \quad (2c)$$

$$V_j^c = \sum_{k=1}^N D_k \frac{W_k}{W} \frac{\partial X_k}{\partial x_j}. \quad (2d)$$

The contributions of the mass diffusion velocity to the total species velocity V_k in Eq. (1d) are usually set to zero. The justification is large turbulent Lewis numbers in coarse mesh simulations, or small mass diffusion compared to other transport processes. Similarly, the explicit viscous effects are neglected in favor of subgrid scale (SGS) modeling of turbulence motions. However, heat conduction effects can significantly affect the flame characteristics, so they are taken into account in detailed SNe Ia calculations. The radiative contribution to the conduction coefficient (λ) can be derived from opacities corresponding to a gray approximation [87] or by more detailed radiative transfer calculations [75]. In comparison with terrestrial combustion, radiation is effectively diffusive at the scales of SNe Ia simulations and hence is lumped into conduction. In stellar interiors, the contribution of electrons to the total energy transport can be neglected compared to that of photons, since conductivity is proportional to the mean free path ℓ , and in non-degenerate stellar material $\ell_{photon} \gg \ell_{particle}$. Nevertheless, conduction by electrons becomes important in the dense degenerate regions in the deep interior of evolved stars, as well as in white dwarfs. In the presence of degeneracy, all quantum states in phase space below Fermi momentum are filled up, and electrons, when approaching ions and other electrons, have difficulty exchanging momentum. This is equivalent to stating that “encounters” are rare or that the mean free path is large [88]. The total conduction coefficient can then be defined in terms of radiation and electron conduction contributions as:

$$\frac{1}{\lambda} = \frac{1}{\lambda_{cond}} + \frac{1}{\lambda_{rad}}, \quad (3)$$

Here, λ_{rad} is defined under equilibrium diffusion approximation as the Rosseland mean of the radiative opacity and λ_{cd} is the electron conduction coefficient written as a “conductive opacity” [88].

In Eq. (2c), B_k is the binding energy (equivalent to the intensive species enthalpy) of the k^{th} species and is calculated as:

$$B_k = \frac{\partial \mathcal{F}}{\partial Y_k} \Big|_{\rho, T} - T \frac{\partial}{\partial T} \left(\frac{\partial \mathcal{F}}{\partial Y_k} \Big|_{\rho, T} \right)_{p, Y_k}, \quad (4)$$

where $\mathcal{F} = \mathcal{F}(\rho, T, Y_k)$ is the Helmholtz free energy. As explained

below, it is convenient to give the equation of state as tables of Helmholtz free energy and its derivatives. With $\mathcal{F}$ specified, other thermodynamic properties, such as the internal energy, $e = \mathcal{F} - T \frac{\partial \mathcal{F}}{\partial T}$, can be subsequently calculated.

2.2. Equations of state

Models of stellar events require the relationship between various thermodynamic properties over a large span of temperatures, densities, and compositions. In highly degenerate matter, the largest pressure contribution comes from electrons and positrons, although the ion and radiation contributions are generally non-negligible. Due to the large number of calls to equations of state (EOS) routines in hydrodynamic models of stellar phenomena, it is desirable to have an EOS model that is efficient and accurately represents the relevant physics. The accuracy, thermodynamic consistency, and execution speed of several EOS routines [27,89–92] which have been used in Type Ia supernova calculations are analyzed by Timmes & Arnett [93]. These EOSs represent improvements in terms of accuracy over a wider range of conditions compared to the earlier versions (e.g. Refs. [94,95]) also listed in Table 1. An electron-positron EOS based on table interpolation of the Helmholtz free energy $\mathcal{F}(\rho, T)$ was developed and analyzed by Timmes & Swesty [96]. The interpolation scheme guarantees thermodynamic consistency to machine precision in IEEE double precision arithmetic, independent of the chosen interpolating function. The choice of a bi-quintic Hermite polynomial as the interpolating function results in accurate reproduction of the underlying Helmholtz free energy and yields smooth derivatives of the pressure, specific entropy, and specific internal energy. The Helmholtz EOS is approximated such that simple multiplication or division by the composition Y_e variables (see definition below) in the appropriate places gives the desired composition scaling. In this way, the tables are two-dimensional and can be used for any composition. The convenience of having two-dimensional tables is also the main reason not to include ion and radiation contributions to the table formulations. Thanks to Frank Timmes, the Nadyozhin, Helmholtz, and some other EOS routines, are available in FORTRAN codes online.¹ The input for these routines includes the temperature, density, the mean number of nucleons per isotope, and the mean charge per isotope. The output includes the total pressure (p_{tot}), the total specific internal energy (e_{tot}), and total specific entropy (s_{tot}).

For the model description, let isotope k have Z_k protons and A_k nucleons (protons + neutrons). Let the aggregate total of isotope k have a mass density ρ_k and a number density n_k in a material with the temperature T (in K) and the total mass density ρ . The mass fraction of isotope k is defined as $Y_k = \rho_k / \rho = n_k A_k / \rho N_A$ where $N_A = 6.02252 \times 10^{26}$ particles/kmol is Avogadro's number. The mean number of nucleons per isotope is $A_m = (\sum Y_k / A_k)^{-1}$, and is the equivalent of the mixture molar mass from the combustion literature [97]. The mean charge per isotope is $Z_m = A_m \sum Z_k Y_k / A_k$, and the number of electrons per baryon (which are the usual matter particles) is $Y_e = Z_m / A_m$. The total scalar pressure is denoted by p_{tot} , total specific internal energy by e_{tot} , and total specific entropy by s_{tot} . Other quantities such as the specific heats or adiabatic indices can be determined once the partial derivatives of the pressure and specific internal energy with respect to the density and temperature are known. The equation of state most often used in SNe Ia calculations follow the description in Ref. [98]. In this EOS, the contributions from different fundamental particles (ions, photons, electrons, and positrons) are considered additive. That is, for instance, for the pressure:

$$p_{tot} = p_{ion} + p_{rad} + p_{ele} + p_{pos}, \quad (5a)$$

where the subscripts “ion,” “rad,” “ele,” and “pos” represent the contributions from nuclei (ions), radiation, electrons, and positrons,

¹ http://cococubed.asu.edu/code_pages/eos.shtml.

Table 1
Computer codes for simulation of SN Ia.

First Author/Code Name	Method	Order	Turbulence Treat.	Flame Treat.	Dim.
Khokhlov et al. [3,147]	• Lagrangian • NRN: Based on [3] • EOS: Nadyozhin	?	–	–	1
Livne et al. [94,153,154] (VULCAN)	• Implicit/Explicit von-Neumann Difference • Fully Compressible • NRN: Based on [2] • EOS: Based on [94]	1	–	Volume of Fluid (VOF)	2
García-Senz et al. [134,155–157]	• Smooth Particle Hydrodynamics (SPH) • NRN: Based on [115] • EOS: Nadyozhin as described in Ref. [91]	0	–	Flame Tracking	3
Gamezo et al. [116,158,159]	• Finite Volume (FV) Adaptive Mesh Refinement (AMR) + Godunov • Fully Compressible NRN: Based on [3] • EOS: Nadyozhin	2	ILES	Flame Capturing	3
Aspdén et al. [103,160–164]	• FV (AMR + PPM) Subcycling in Time • Low M Approx. • NRN: 7 Species • EOS: Helmholtz	2	ILES	–	3
Almgren et al. [102, 165–167] (MAESTRO)	• FV AMR + Piecewise Parabolic Method (PPM) • Low M Approx. • NRN: 7/13/19 Species • EOS: Helmholtz	2	ILES	–	3
Almgren et al. [101,168] (CASTRO)	• FV AMR + PPM • Fully Compressible • NRN: 7/13/19 Species • EOS: Helmholtz	2	ILES	Thickened Flame Model (TFM) in Ref. [55]	3
Reinecke et al. [114,139,169–171] (PROMETHEUS)	• FV PPM • Fully Compressible • NRN: Based on [114] • EOS: Based on [95]	2	One-equation SGS viscosity LES	Level Set Method (LST)	3
Röpke et al. [172–179] (PROMETHEUS)	• FV PPM + Moving Mesh • Fully Compressible • NRN: Based on [114] • EOS: Based on [95]	2	One-equation SGS viscosity LES	LST + Multi-Grid	3
Fryxell et al. [100,180–182] (FLASH: Based on PROMETHEUS)	• FV AMR + PPM • Fully Compressible • NRN: 7/13/19 Species • EOS: Helmholtz	2	ILES	Flame Capturing	3
Seitenzahl et al. [61, 75, 136, 183] (LEAF: Based on PROMETHEUS)	• FV PPM + Moving Mesh • Fully Compressible • NRN: Based on [114] • EOS: Based on [95]	2	One-equation SGS viscosity LES	LST + Multi-Grid	3

respectively. The contribution from Coulomb corrections to the equation of state can be added following Ref. [99], although it is usually neglected for degenerate matter in high-temperature regimes.

Ions: In the deep interiors of stars, the materials are fully ionized. We therefore have a mixture of nuclei (which in itself can consist of more than one isotope) and free electrons and positrons. The mixture of nuclei can be treated as a one-component gas, if all single components obey the perfect gas equation. The ion portions of pressure and internal energy corresponding to ideal gas, with $\gamma = 5/3$, and the Sacker-Tetrode equation for the specific entropy typically used in literature [100–102] are:

$$p_{ion} = \frac{N_A k_B}{A_m} \rho T = \mathcal{R} \sum \frac{Y_k}{A_k} \rho T, \quad (6a)$$

$$e_{ion} = \frac{1}{\gamma - 1} \frac{p_{tot}}{\rho}, \quad (6b)$$

$$s_{ion} = \frac{p_{ion}/\rho + e_{ion}}{T} + \frac{N_A k_B}{A_m} \log \left[\frac{A_m^{5/2}}{\rho N_A} \left(\frac{2\pi k_B T}{m_p h^2} \right)^{3/2} \right]. \quad (6c)$$

Radiation: In stellar interiors, the photons and fermions may also contribute noticeably to the thermodynamic properties; therefore their contributions should be added to the equation of state. The conditions are very close to thermal equilibrium, and the radiation very close to that of a black body [88], so the radiation portions of pressure, energy,

and entropy are typically expressed as [88,100,103]:

$$p_{rad} = \frac{aT^4}{3}, \quad e_{rad} = \frac{3p_{rad}}{\rho}, \quad s_{rad} = \frac{(p_{rad}/\rho + e_{rad})}{T}. \quad (7)$$

Electrons and positrons: Analytical relations are available for contributions due to electrons and positrons. Based on the formalism of a non-interacting Fermi gas for the electrons and positrons, the number densities of free electrons, N_{ele} , and positrons, N_{pos} , are:

$$N_{ele} = \frac{8\pi\sqrt{2}}{h^3} m_e^3 c^3 \beta^{3/2} [F_{1/2}(\eta, \beta) + F_{3/2}(\eta, \beta)], \quad (8a)$$

$$N_{pos} = \frac{8\pi\sqrt{2}}{h^3} m_e^3 c^3 \beta^{3/2} [F_{1/2}(-\eta - 2/\beta, \beta) + \beta F_{3/2}(-\eta - 2/\beta, \beta)], \quad (8b)$$

where $\beta = kT/(m_e c^2)$ is the relativity parameter, $\eta = \mu_p/kT$ is the normalized chemical potential of electrons, and $F_k(\eta, \beta)$ is the Fermi-Dirac integral:

$$F_k(\eta, \beta) = \int_0^\infty \frac{x^k (1 + 0.5\beta x)^{1/2}}{1 + e^{(x-\eta)}} dx. \quad (9)$$

For complete ionization, the number density of free electrons in the matter is: $N_{ele,matter} = Z_m N_{ion} = (Z_m/A_m) N_A \rho = Y_e N_A \rho$, and the charge neutrality requires: $N_{ele,matter} = N_{ele} - N_{pos}$. With these, the normalized chemical potential η is calculated. That allows determination of the pressure, the specific internal energy, and the entropy due to the free electrons and positrons.

$$p_{ele} = \frac{16\pi\sqrt{2}}{3h^3} m_e^4 c^5 \beta^{5/2} \left[F_{3/2}(\eta, \beta) + \frac{1}{2\beta F_{5/2}(\eta, \beta)} \right], \quad (10a)$$

$$p_{pos} = \frac{16\pi\sqrt{2}}{3h^3} m_e^4 c^5 \beta^{5/2} \left[F_{3/2}(-\eta - 2/\beta, \beta) + \frac{1}{2\beta F_{5/2}(-\eta - 2/\beta, \beta)} \right], \quad (10b)$$

$$e_{ele} = \frac{8\pi\sqrt{2}}{\rho h^3} m_e^4 c^5 \beta^{5/2} [F_{3/2}(\eta, \beta) + \beta F_{5/2}(\eta, \beta)], \quad (10c)$$

$$e_{pos} = \frac{8\pi\sqrt{2}}{\rho h^3} m_e^4 c^5 \beta^{5/2} [F_{3/2}(-\eta - 2/\beta, \beta) + \beta F_{5/2}(-\eta - 2/\beta, \beta)] + \frac{2m_e c^2 N_{pos}}{\rho}, \quad (10d)$$

$$s_{ele} = \frac{1}{T} \left(\frac{p_{ele}}{\rho} + e_{ele} - \eta k_B T N_a N_{ele} \right), \quad (10e)$$

$$s_{pos} = \frac{1}{T} \left(\frac{p_{pos}}{\rho} + e_{pos} - \eta k_B T N_a N_{pos} \right). \quad (10f)$$

Note that evaluation of the electron-positron portion of the EOS requires root finding of the charge neutrality equation and evaluation of the Fermi-Dirac integrals as given by Eq. (9), which is not computationally efficient and not typically used. Instead, pre-computed tables of the Helmholtz free energy and its derivatives, and reading routines are used [98]. The electron-positron contributions to internal energy and pressure can then be calculated as:

$$e_{ele+pos} = \mathcal{F} - T \frac{\partial \mathcal{F}}{\partial T}, \quad p_{ele+pos} = \rho^2 \frac{\partial \mathcal{F}}{\partial \rho}. \quad (11)$$

Formulating the EOS in terms of Helmholtz free energy ensures thermodynamic consistency [96]. Ref. [98] provides Helmholtz EOS routines covering the range $10^{-7} < \rho < 10^{14}$ (kg m⁻³) and $10^4 < T < 10^{11}$ (K), that can be called pointwise. Since hydrodynamic calculations advance density and internal energy, backward relations ($p_{ele+pos}$, T as functions of ρ , $e_{ele+pos}$, and composition) are also needed. This can be accomplished by performing Newton-Raphson iterations to determine the temperature [100], and then call the forward EOS routines to find the pressure.

2.3. Source terms

In the absence of coupling to neighboring regions through mass diffusion or advection, any changes to the composition are due to local processes. With this assumption, Eq. (1d) is transformed into a set of ODEs, $\dot{Y} = f(Y, \rho, T)$. This set of ODEs forms a “reaction network” (RN). For the reaction

$$\sum_{k=1}^N \nu'_{kj} M_{kj} \rightleftharpoons \sum_{k=1}^N \nu''_{kj} M_{kj}, \quad j = 1, \dots, M_r, \quad (12)$$

the reaction rate for the species k is the sum of rates produced by all M_r reactions:

$$\dot{\omega}_k = \sum_{j=1}^N \dot{\omega}_{kj} = \frac{A_k}{N_A} \sum_{j=1}^N \nu_{kj} Q'_j, \quad (13)$$

where,

$$\nu_{kj} = \nu''_{kj} - \nu'_{kj}, \quad (14a)$$

$$Q'_j = K_{f,j} \prod_{k=1}^N (n_k)^{\nu'_{kj}} - K_{r,j} \prod_{k=1}^N (n_k)^{\nu''_{kj}} \quad (14b)$$

$$= K_{f,j} \prod_{k=1}^N \left(\frac{\rho Y_k N_A}{A_k} \right)^{\nu'_{kj}} - K_{r,j} \prod_{k=1}^N \left(\frac{\rho Y_k N_A}{A_k} \right)^{\nu''_{kj}},$$

$$\dot{Q} = \sum_{j=1}^N Q'_j S_j. \quad (14c)$$

The quantity $\prod_{k=1}^N (n_k)^{\nu'_{kj}}$ is the total possible number of elementary reactions per unit volume (obtained by counting the number of possible collisions) and is based on the number density of particles. Not all collisions have the required energy to break the energy barrier, so this needs to be multiplied by the reaction cross section. In combustion, the cross sections (rates of reaction) are usually given by the Arrhenius law, in which experimental data are fit by exponential variations [104]. At SNe Ia densities, the Coulomb logarithm barrier needed to be overcome during ion collisions is greatly reduced by the presence of free electrons. This translates into an exponential screening factor which is necessary to correct reaction rates obtained from standard libraries (see below) and significantly increases the temperature nonlinearity.

The coefficients that appear in the expressions above span many orders of magnitude because the nuclear reaction rates have highly non-linear dependence on the temperature and the abundance levels themselves typically range over several orders of magnitude. As a result, the differential equations that comprise a nuclear RN are stiff. From the physical standpoint, this implies that at least one of the isotopic abundances is changing on a much faster timescale compared to other isotopes.

In terms of CPU time and memory, detailed treatment of nuclear RNs remains very expensive, even after the species partial differential transport equations are reduced to ODEs. The largest portion of the memory is usually reserved for storing the values of isotopic abundance levels at each grid point. This is quite restrictive, especially for 3D simulations. Various methods of solving stiff ODEs are reviewed by Timmes [105]. However, even with modern methods for solving stiff ODEs, evolving the isotopic abundances dominates the total cost of a multi-dimensional model when the number of isotopes evolved is ~ 30 [100,106]. To decrease the computational requirements, a trade-off between the number of isotopes and the spatial resolution is needed. The general response to this problem has been to evolve a reduced number of isotopes, which results in an approximation for thermonuclear energy generation rates and some key isotopic abundances during hydrodynamical explosion simulations, and then to use a nucleosynthesis post-processing step with a large number of species NRN for determining the detailed chemical structure of ejected materials [22]. Sensitivity analysis supports this approach, unless detailed element abundance is targeted, as it shows that the element production and associated derived quantities like luminosity mainly depend on flame conditions and propagation mechanisms rather than small changes in reaction rates [107]. The set of 13 nuclei most commonly used for this purpose is composed of (α, γ) and (γ, α) links² between ^{12}C , ^{16}O , ^{20}Ne , ^{24}Mg , ^{28}Si , ^{32}S , ^{36}Ar , ^{40}Ca , ^{44}Ti , ^{48}Cr , ^{52}Fe , and ^{56}Ni isotopes, (α, p)(p, γ) and (γ, p)(p, α) links through 8 intermediate isotopes (^{27}Al , ^{31}P , ^{35}Cl , ^{39}K , ^{43}Sc , ^{47}V , ^{51}Mn , and ^{55}Co) with steady state proton flow assumption,³ and heavy-ion reactions ($^{12}\text{C} + ^{12}\text{C}$, $^{12}\text{C} + ^{16}\text{O}$, and $^{16}\text{O} + ^{16}\text{O}$). This minimal set of nuclei, usually called an α -chain network, can reasonably track the abundance levels from helium burning through nuclear statistical equilibrium. Moreover, in comparison with much larger nuclear RNs, α -chain RN has $\sim 30\%$ error in calculating the full thermonuclear energy generation rates [106]. In this way, one gets most of the energy generated for a reasonable range of thermodynamic conditions at a fraction of the computational cost. Using any α -chain based RN is reasonably accurate as long as Υ_e is not significantly different from 0.5, and the flow in neutron and proton channels is not substantial [106].

Libraries of thermonuclear reaction rates like JINA REACLIB [108],

² $A(b,c)D$ is equivalent to $A + b \rightleftharpoons c + D$.

³ Here α, γ , and p are representing the alpha particle, photon, and proton, respectively.

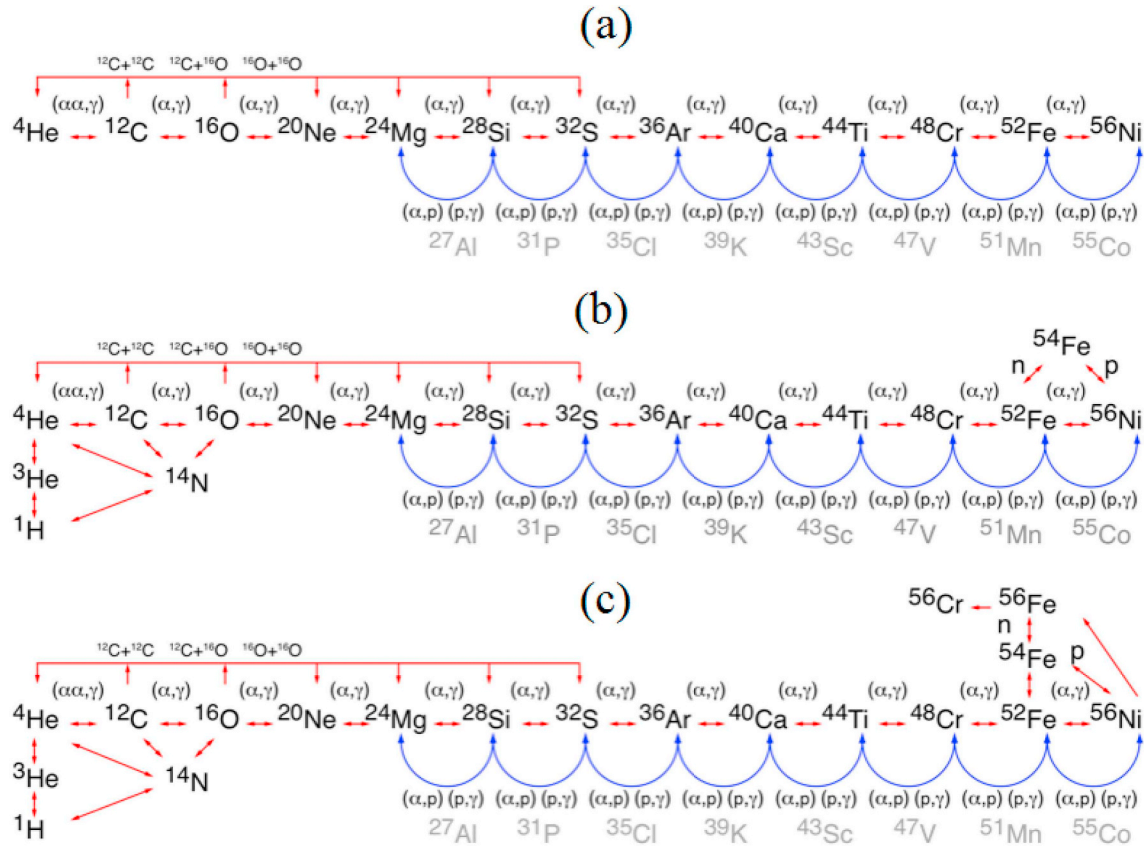


Fig. 2. Reaction networks with (a) 13, (b) 19 and (c) 21 isotopes. Reprinted from Ref. [117] with permission.

STARLIB [109], and Fowler et al. [110–113] can be used for detailed RNs calculations. Reduced RNs with 13, 19, 21, or 34 isotopes are also publicly available as Fortran or Python subroutines.⁴ Compared with the 13 isotope RN, the 19 isotope RN adds some isotopes to accommodate hydrogen burning and photo-disintegration (Fig. 2). Networks with higher number of isotopes relax the assumptions used in the 13 and 19 isotope RNs [105]. Several studies have also used RNs with less than 13 isotopes, e.g. a 7-isotope RN and a 3-stage RN. The former can calculate the energy generation rate relatively accurate for helium, carbon, neon, and oxygen burning [106]. The latter reduces the number of reactions by comparing the reaction time-scales and assuming nuclear statistical equilibrium (NSE) in certain ranges of temperature to further simplify the RN [2,3,114–116].

When using these networks, the conservation of nuclear species $\sum_{k=1}^N X_k = 1$ must be carefully monitored. Failure to satisfy this relation to machine precision may manifest itself in the non-physical buildup (or decay) of the abundances or energy generation rate.

3. Simulations

For stellar conditions typical of C+O flames, the Lewis number is $O(10^7)$, the Prandtl number is $O(10^{-5})$ [4,118], and the characteristic Reynolds number is $O(10^{14})$ [119]. Though the Prandtl number estimates from Ref. [118] are usually assumed to hold for SNe Ia, the analysis might not be applicable to the progenitor conditions. Nevertheless, for these parameter values, obviously direct numerical simulation (DNS) of full scale SN flames is totally out of question. The situation will surely remain the same even with the next generation of exascale supercomputing platforms. In most simulations to date, the inviscid forms of the equations are considered (Euler hydrodynamics),

and the primary source of dissipation to regularize the equations is furnished implicitly by the numerical algorithm. In general, these calculations are denoted as implicit large eddy simulation (ILES), even though this description encompasses very different numerical algorithms with qualitatively different numerical properties. In some cases, the control of this dissipation is provided by adaptive mesh refinement (AMR). Such a generic description sometimes can cause confusion in interpreting the results, thus affecting the reproducibility of the simulations via different algorithms. Table 1 provides a listing of several of the more popular numerical approaches with the corresponding computer codes for simulating SNe Ia. The majority of these codes are based on conventional low-order finite volume (FV) schemes; some of them including simple SGS closures.

Most of the earlier SN explosion investigations have been via “classical” one-dimensional (1D) spherically symmetric models. The deflagration W7 model of Nomoto et al. [70] has successfully explained the basic contribution of SNe Ia to the galactic chemical evolution, as well as some basic features of the spectra and light curves of individual SNe Ia of a normal class [120]. This model, however, cannot capture many features of multidimensional thermonuclear burning for M_{Ch} explosion, such as off-center flame ignition, flame instabilities, and turbulence [38,74,121]. With increased availability of supercomputers, simulations have been extended to 2D [2,94,122–132], and 3D [61,133–136]. Along with these simulations come significant developments of numerical algorithms for representing thin propagating surfaces [5,137–139]. The results of these multidimensional simulations have been useful in elucidating some of the fundamental issues in SN explosions [75,140–143]. These studies suggest that SNe Ia are the outcome of several possible explosion mechanisms after a variety of evolutionary paths towards the explosion [50]. These paths result in different scenarios for ^{56}Ni yield, spatial distribution of the SN remnant, kinetic energy, expansion speed, and gamma-ray line light curves. As SNe Ia are the main producers of iron-peak elements (Cr, Mn, Fe, Co

⁴ http://cococubed.asu.edu/code_pages/burn_helium.shtml.

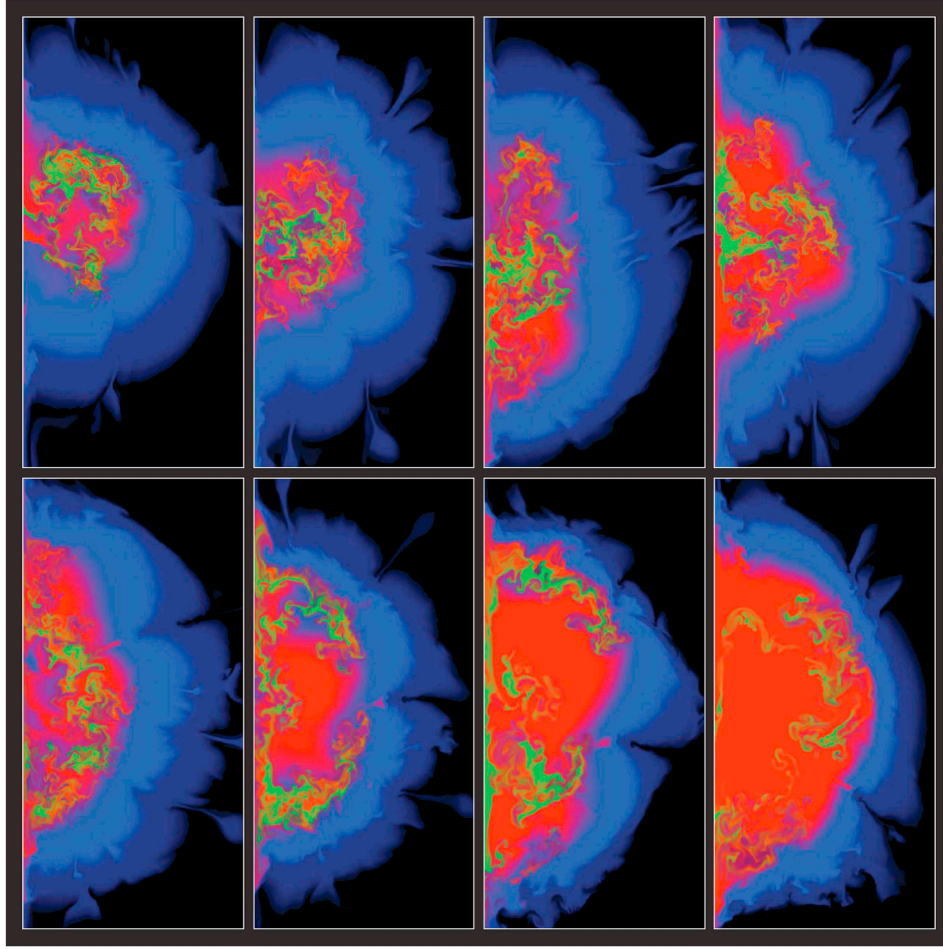


Fig. 3. Chemical structure of the SN Ia ejected debris for a subset of explosion models. Blue, intermediate-mass elements (that is, silicon, sulphur, calcium); green, stable iron-group elements produced by electron capture; and red, ^{56}Ni . Reprinted from Ref. [33] with permission.

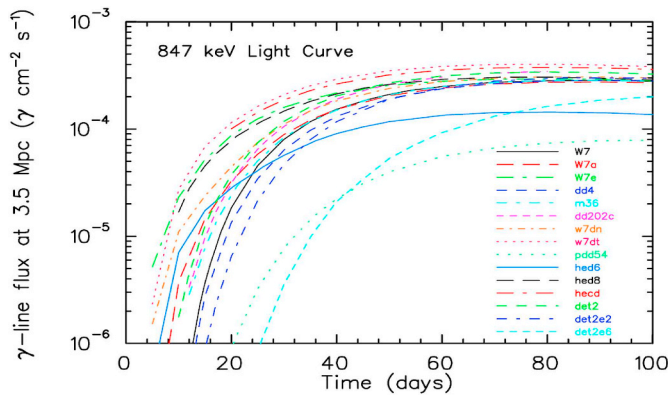


Fig. 4. Light curves in photons $\text{cm}^{-2}\text{s}^{-1}$ vs. time (in days after explosion) of the 847 keV line of ^{56}Co decay at a distance of 3.5 Mpc for 15 SNe Ia models. The peak emission occurs between day 60 and day 100, but the time at which this light curve first exceeds a given threshold is a sharp function of model. The fluxes at day 40, for instance, vary by almost an order of magnitude, and the fluxes at day 20 can vary by a factor of 30. Reprinted from Ref. [81] with permission.

and Ni) and their ejecta affects the chemical evolution of galaxies, new simulations with refined meshes and better physics by using advanced SGS models are needed for clarifying the detailed physics and probabilities of different explosion paths, and also identifying the role of other mechanisms such as deflagration to detonation transition and turbulence [82,144].

To exemplify the complexities associated with predicting the amount and distribution of ^{56}Ni , Fig. 3 shows the chemical structure of the ejected debris 100s after ignition for a subset of the explosion models with different ignition and detonation conditions [33]. The ^{56}Ni production increases (left to right) for models which undergo relatively more burning in the detonation as compared to the deflagration phase of the explosion. The turbulent inner regions reflect Rayleigh-Taylor and other instabilities that develop during the initial deflagration phase of burning. The subsequent detonation wave enhances the ^{56}Ni production in the center by burning remaining pockets of fuel. The lower-density outer layers of debris, processed only by the detonation, consist of smoothly distributed intermediate-mass elements. Due to the sensitivity of the ^{56}Ni distribution to the details of the model, it is hoped that comparisons with new observations can help assess and improve the physical and computational models of SNe Ia.

Another feature that distinguishes different models is their predictions of the light curves. Fig. 4 shows the expected light curves (in days after explosion) of the 847 keV line of ^{56}Co decay at a distance of 3.5 Mpc from earth for a set of 15 explosion models for SN2014J. This is the closest SN Ia that has occurred within the past 40 years, so that the gamma-ray spectrometer on ESA's *INTEGRAL* gamma-ray observatory [145] could measure its gamma-ray lines. The expected results in Fig. 4 span the model space discussed over the last 30 years containing pure deflagration (w7, w7a, and w7e all from Ref. [38]), delayed detonation (dd4 from Ref. [146], w7dn and w7dt both from Ref. [76], m36 from Ref. [147], and dd202c from Ref. [148]), merger detonation (det2, det2e2, and det2e6 from Refs. [78,149]), pulsational delayed detonation (pdd54 from Ref. [150]), and sub-Chandrasekhar helium

detonation (hed6, hed8, and hecd from Ref. [78]). The elements and density distributions used for generating these curves are extracted from the original models described in the literature; for example, w7 model assumes a WD with the total mass of $1.38 M_{\odot}$, $0.58 M_{\odot}$ ^{56}Ni yield, and 10^{51} ergs explosion energy. In all models, the density and species distributions are assumed to become homogeneous early during SN expansion. The peak emission occurs for most models approximately between day 60 and day 100, but the early rise of the light curve is a stiff function of model [81]. By comparing the model predictions with the observations, Ref. [151] states that the whole class of near-Chandrasekhar models provides a reasonable description of the data, with the pure deflagration and delayed detonation models being the most successful. However, Ref. [152] makes the point that one cannot discriminate among the predicting models because the observed line shapes clearly suggest the explosion is not spherically symmetric.

4. The prospects of large eddy simulation

With the largest numerical resolution that can be currently afforded for SN simulations, the SGS energy ignored due to lack of resolution constitutes a significant portion of the total energy. Large eddy simulation (LES) is a formal means of accounting for the effects of these scales [184,185]. The importance of LES has been well-recognized within the astrophysics community [186,187]. Such simulations provide the optimum means of turbulence predictions, but require the input of reliable SGS closures. Within the past 30 years, LES has proven to be a very effective tool for turbulent combustion predictions. We feel that it is now the time to extend it more systematically for SN simulations.

Implementation of LES involves the use of the spatial filtering operation [188,189] $\bar{Q}(\mathbf{x}, t) = \int_{-\infty}^{+\infty} Q(\mathbf{x}', t) \mathcal{H}(\mathbf{x}', \mathbf{x}) d\mathbf{x}'$, where $\mathcal{H}$ denotes the filter function of width $\Delta_{\mathcal{H}}$, and $\bar{Q}(\mathbf{x}, t)$ represents the filtered value of the transport variable $Q(\mathbf{x}, t)$. In compressible turbulent flows [190], it is convenient to consider the Favré filtered quantity, $\bar{Q}(\mathbf{x}, t) = \bar{\rho} \bar{Q} / \bar{\rho}$. With application of a spatially invariant and localized filter function, the filtered forms of the equations given above are:

$$\frac{\partial \bar{\rho}}{\partial t} + \frac{\partial \bar{\rho} \bar{u}_j}{\partial x_j} = 0, \quad (15a)$$

$$\frac{\partial \bar{\rho} \bar{u}_i}{\partial t} + \frac{\partial \bar{\rho} \bar{u}_i \bar{u}_j}{\partial x_j} = -\frac{\partial}{\partial x_j} [\bar{p}_{tot} \delta_{ij} - \bar{\tau}_{ij} - \bar{\tau}_{ij}^t] + \sum_{k=1}^N \frac{\partial}{\partial x_j} [\bar{\rho} Y_{k,i} \bar{f}_{k,i}], \quad (15b)$$

$$\begin{aligned} \frac{\partial \bar{\rho} \bar{E}_{tot}}{\partial t} + \frac{\partial \bar{\rho} \bar{E}_{tot} \bar{u}_j}{\partial x_j} = & -\frac{\partial}{\partial x_j} \left[\bar{u}_i (\bar{p}_{tot} \delta_{ij} - \bar{\tau}_{ij}) + \bar{q}_j + \bar{q}_j^t \right] + \bar{Q} \\ & + \sum_{k=1}^N \frac{\partial}{\partial x_j} [\bar{\rho} Y_{k,i} \bar{f}_{k,i} (\bar{u}_i + V_{k,i})], \end{aligned} \quad (15c)$$

$$\frac{\partial \bar{\rho} \bar{Y}_k}{\partial t} + \frac{\partial \bar{\rho} \bar{Y}_k \bar{u}_j}{\partial x_j} = -\frac{\partial}{\partial x_j} [\bar{J}_{j,k} + \bar{J}_{j,k}^t] + \bar{\omega}_k. \quad (15d)$$

The transport terms usually contain their own non-linearities, which can significantly increase the modeling difficulty. In most combustion studies, the correlations between the transport coefficients and the primary variables are neglected:

$$\bar{\tau}_{ij} = 2\mu \left(\bar{\mathcal{S}}_{ij} - \frac{1}{3} \bar{\mathcal{S}}_{ij} \bar{\mathcal{S}}_{ll} \right) \approx 2\mu \left(\bar{\mathcal{S}}_{ij} - \frac{1}{3} \bar{\mathcal{S}}_{ij} \bar{\mathcal{S}}_{ll} \right), \quad (16a)$$

$$\bar{\mathcal{S}}_{ij} = \frac{1}{2} \left(\frac{\partial \bar{u}_j}{\partial x_i} + \frac{\partial \bar{u}_i}{\partial x_j} \right), \quad (16b)$$

$$\bar{J}_{j,k} \approx -\rho \left(D_k \frac{W_k}{W} \frac{\partial \bar{X}_k}{\partial x_j} - Y_k V_j^c \right) \approx -\bar{\rho} \left(D_k \frac{W_k}{W} \frac{\partial \bar{X}_k}{\partial x_j} - \bar{Y}_k \bar{V}_j^c \right), \quad (16c)$$

$$\bar{q}_j \approx -\bar{\lambda} \frac{\partial \bar{T}}{\partial x_j} + \sum_{k=1}^N \bar{J}_{j,k} \bar{\tilde{B}}_k. \quad (16d)$$

Such simplifications are fully justified in two opposing scenarios. First, when the cut-off filter is close to the viscous/diffusive scales, the SGS fluctuations are small enough that the transport terms can be approximated by expressions involving only filtered sub-terms. However, for SNe Ia calculations, the second scenario applies, mainly the grid sizes are coarse enough that the turbulent transport is much larger than the viscous/diffusive counterparts. The heat flux though is likely to remain important compared to the turbulent energy flux even on the scale of the current SNe Ia simulations. Further research is needed to assess if approximation (16d) is reasonable, or the correlation between λ and temperature gradient needs to be explicitly considered. In the latter case, one would need to model the SGS flux $\left(-\bar{\lambda} \frac{\partial \bar{T}}{\partial x_j} + \bar{\lambda} \frac{\partial \bar{T}}{\partial x_j} \right)$. The filtered pressure and energy terms also contain higher order non-linearities:

$$\bar{p}_{rad} = \frac{a \bar{T}^4}{3} = \frac{a \bar{T}^4}{3} + \left(\frac{a \bar{T}^4}{3} - \frac{a \bar{T}^4}{3} \right), \quad (17a)$$

$$\bar{p}_{ion} = \frac{N_A k_B}{A_m} \bar{\rho} \bar{T}, \quad (17b)$$

$$\bar{p}_{ele} = \frac{16\pi\sqrt{2}}{3h^3} m_e^4 c^5 \beta^{5/2} \left[F_{3/2}(\eta, \beta) + \frac{1}{2\beta F_{5/2}(\eta, \beta)} \right], \quad (17c)$$

$$\bar{p}_{pos} = \frac{16\pi\sqrt{2}}{3h^3} m_e^4 c^5 \beta^{5/2} \left[F_{3/2}(-\eta - 2/\beta, \beta) + \frac{1}{2\beta F_{5/2}(-\eta - 2/\beta, \beta)} \right], \quad (17d)$$

$$\bar{e}_{rad} = \frac{3\bar{p}_{rad}}{\bar{\rho}}, \quad (17e)$$

$$\bar{e}_{ion} = \frac{1}{\gamma - 1} \frac{\bar{p}_{tot}}{\bar{\rho}}, \quad (17f)$$

$$\bar{e}_{ele} = \frac{1}{\bar{\rho}} \left[\frac{8\pi\sqrt{2}}{h^3} m_e^4 c^5 \beta^{5/2} [F_{3/2}(\eta, \beta) + \beta F_{5/2}(\eta, \beta)] \right], \quad (17g)$$

$$\bar{e}_{pos} = \frac{1}{\bar{\rho}} \left[\frac{8\pi\sqrt{2}}{h^3} m_e^4 c^5 \beta^{5/2} [F_{3/2}(-\eta - 2/\beta, \beta) + \beta F_{5/2}(-\eta - 2/\beta, \beta)] \right] \quad (17h)$$

$$+ \frac{1}{\bar{\rho}} \left[\frac{2m_e c^2 N_{pos}}{\bar{\rho}} \right],$$

where

$$\bar{p}_{tot} = \bar{p}_{rad} + \bar{p}_{ion} + \bar{p}_{ele} + \bar{p}_{pos}, \quad (18a)$$

$$\bar{e}_{tot} = \bar{e}_{rad} + \bar{e}_{ion} + \bar{e}_{ele} + \bar{e}_{pos}. \quad (18b)$$

Except for the simple ideal gas equation of state, which is already arranged in a form appropriate for Favre filtering, the rest of the contributions to the total pressure and internal energy require considerations of the SGS terms. These contributions and the associated filtered quantities also need to satisfy thermodynamic consistency conditions. Unfortunately, there is very little work in the LES literature on modeling of such quantities. Probability density function (PDF) modeling techniques (discussed below), in which the statistics of the primary variables are advanced, could, in principle, address these issues and suggest appropriate reduced order modeling approaches. High fidelity resolved simulations are also needed to test the modeling ideas. The prospects for such simulations are discussed in §5. The SGS closure problem involves all of the terms presented i.e. in Eqs. (16) and (17):

$$\bar{\tau}_{ij}^t = -\bar{\rho} (\bar{u}_i \bar{u}_j - \bar{u}_i \bar{u}_j), \quad (19a)$$

Table 2
Turbulence and flame treatment methods used in SN Ia simulations.

SGS Methodology	SGS-Flame Model
• Implicit Large Eddy Simulation (ILES) [103,160–164] • Explicit Large Eddy Simulation (LES) One-equation SGS viscosity [61,75,114,136,170–179,183] • LEM [199,200]	• Flame Capturing [100,116,158,159,180–182] • Flame Tracking [134,155–157] • Level Set Technique (LST) [61,75,114,136,139,169–179,183] • Volume of Fluid (VOF) [94,153,154]

$$\bar{J}_{j,k}^t = \bar{\rho}(\bar{u}_j \bar{Y}_k - \bar{u}_j \bar{Y}_k), \quad (19b)$$

$$\bar{q}_j^t = \bar{\rho}(\bar{u}_j \bar{E}_{tot} - \bar{u}_j \bar{E}_{tot}), \quad (19c)$$

$$\bar{\omega}_k = \sum_{k=1}^N \bar{\omega}_{kj} = \frac{A_k}{N_A} \sum_{k=1}^N \nu_{kj} \bar{Q}'_j, \quad (19d)$$

$$= \frac{A_k}{N_A} \sum_{k=1}^N \nu_{kj} \left[\overline{K_{f,j} \prod_{k=1}^N (n_k)^{\nu'_{kj}}} - \overline{K_{r,j} \prod_{k=1}^N (n_k)^{\nu''_{kj}}} \right]$$

$$\approx \frac{A_k}{N_A} \sum_{k=1}^N \nu_{kj} \left[\overline{K_{f,j}} \prod_{k=1}^N \left(\frac{\bar{\rho} \bar{Y}_k N_A}{A_k} \right)^{\nu'_{kj}} - \overline{K_{r,j}} \prod_{k=1}^N \left(\frac{\bar{\rho} \bar{Y}_k N_A}{A_k} \right)^{\nu''_{kj}} \right],$$

$$\bar{Q} = \sum_{k=1}^N \bar{Q}'_j \bar{S}_j \approx \sum_{k=1}^N \bar{Q}'_j S_j \quad (19e)$$

$$\approx \sum_{k=1}^N S_j \left[\overline{K_{f,j}} \prod_{k=1}^N \left(\frac{\bar{\rho} \bar{Y}_k N_A}{A_k} \right)^{\nu'_{kj}} - \overline{K_{r,j}} \prod_{k=1}^N \left(\frac{\bar{\rho} \bar{Y}_k N_A}{A_k} \right)^{\nu''_{kj}} \right].$$

Significant progress has been made within the past 30 years to account for modeling of the SGS terms defined by Eq. (19). Most of the models currently in use originate from the traditional Reynolds-averaging methodology, commonly referred to as Reynolds-averaged Navier-Stokes (RANS) [191,192]. Table (2) provides a listing of such models as currently used for SN simulations. These constitute a very small portion of the closures in turbulent combustion [184,193–198]. Here, a review is provided of these closures with a speculation of their future implementation for SNe Ia predictions.

No-Model: With this approach, all of SGS contributions are assumed negligible and are totally neglected. In a version of this approach, termed implicit large eddy simulation (ILES), justification is made by arguing that the numerical algorithm provides all of the

necessary SGS closure [161,201,202]. In non-reactive flows, the approach sometimes performs well as the discretized scheme portrays the cascade character of the energy for specific numerical algorithms. It appears that Schumann [203] was one of the first to conduct LES of reacting flows in this manner. Since then, no-model LES has been employed in both turbulent combustion [204–206], and turbulent thermo-nuclear reactions [162,163] (Fig. 5). However, it cannot be trusted for general applications, because it can lead to significant errors in LES of reacting flows. For such flows, the contributions of SGS scalar fluctuations to the filtered reaction rate are substantial. This has been long recognized in RANS in both combustion [207] and chemical engineering [208,209]. In addition, different numerical algorithms generally have qualitatively different dissipative properties, which affects the reproducibility of the results.

Boussinesq-Type Approximation: This class of models is based on the original idea of Smagorinsky's [210] SGS viscosity closure, which relates the unknown SGS Reynolds stresses to the local large scale rate of flow strain [211]. This viscosity aims to mimic the dissipative behavior of the unresolved small scales. The extensions to “dynamic” models [212,213], using structural tensors [214], and near wall models [215,216] have shown some improvements. This is particularly the case in transitional flow simulations, where the dynamic evolution of the empirical model “constant” results in better predictions of the large scale flow features [217,218]. However, the model cannot account for SGS turbulence-reaction kinetics interactions. Further extensions of the model relate the SGS viscosity to the SGS kinetic energy, which is determined from a dynamical equation. Several successful examples of SNe Ia calculations using a one-equation SGS viscosity model are reported in Refs. [61,75,114,136,170–179,183], and are exemplified in Figs. 6 and 7. In particular, Fig. 6 shows that the SGS transport effects (right bottom panels) are important in the vicinity of the flame front. This transport cannot be captured by the algebraic Smagorinski type models.

Moment Methods: The obvious extension of the no-model

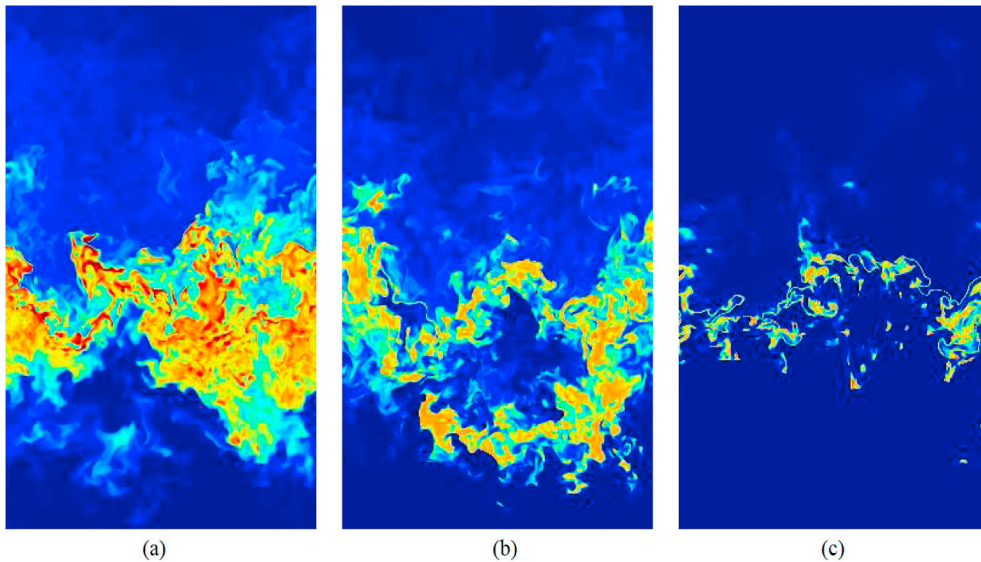


Fig. 5. Instantaneous vertical slices of fuel consumption rate in a SN Ia simulation (intense burning in red, no burning in blue). As the flame burns in increasingly larger integral length scales, there is a decrease in the relative size of the flame structures, with burning occurring in scattered pockets. Periodic boundary conditions are prescribed laterally in this simulation. The whole lateral domain width is shown, but the height is cropped. The widths of the three calculations are 150 cm, 1200 cm, and 9600 cm. Reprinted from Ref. [163] with permission.

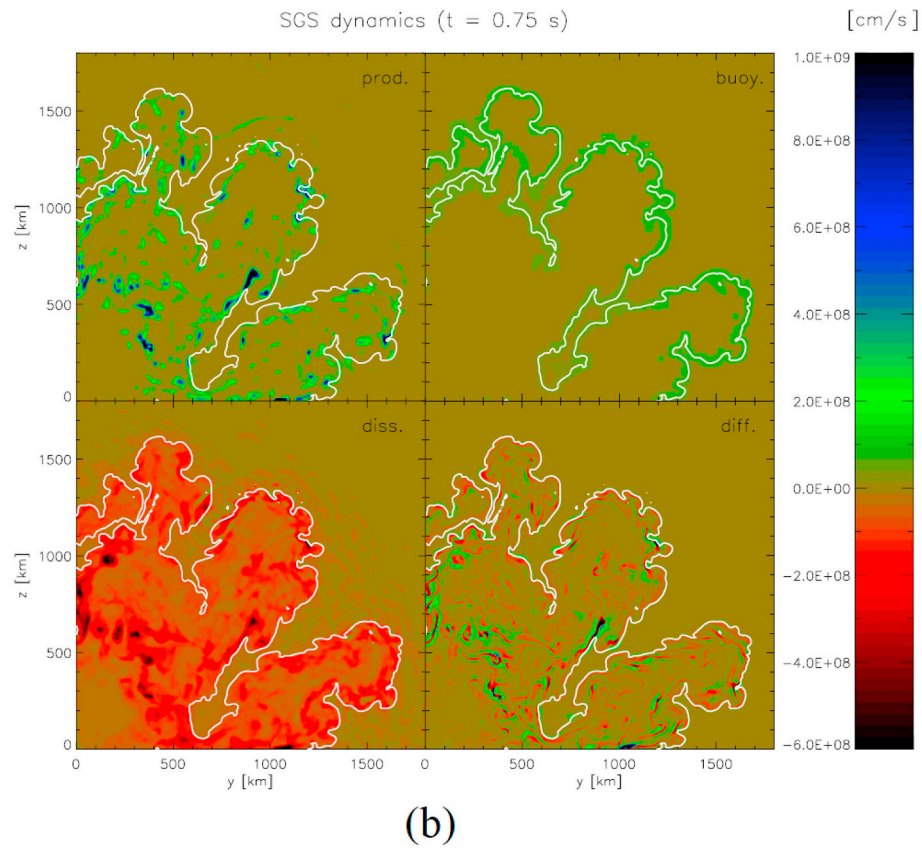
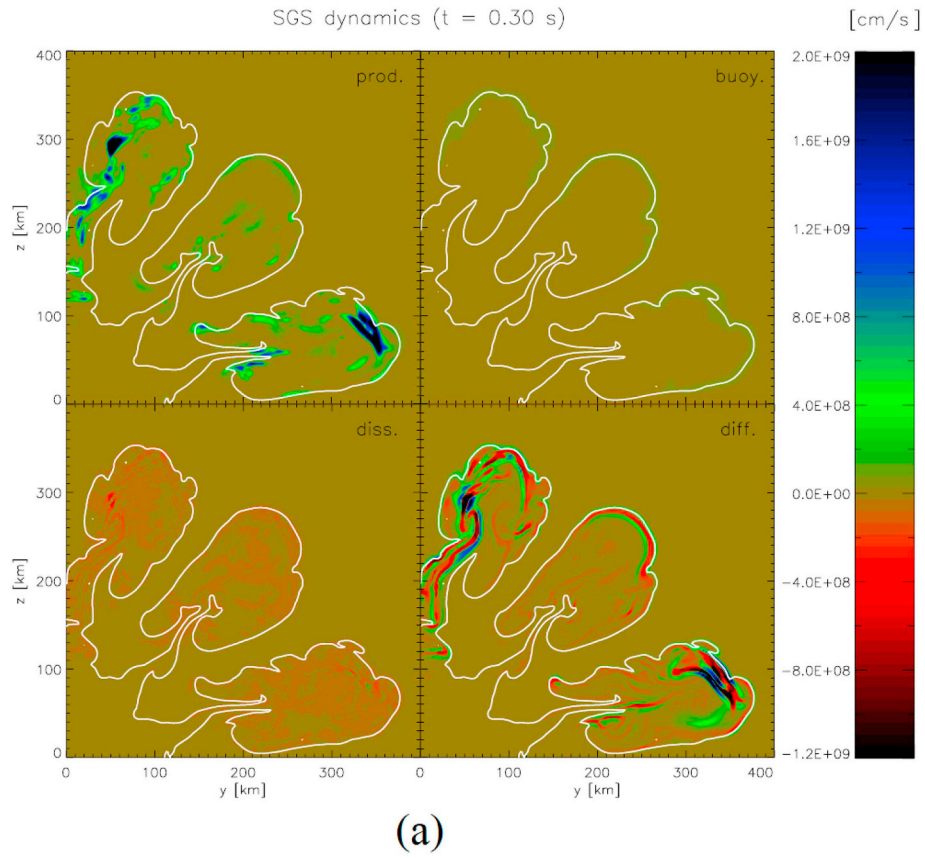


Fig. 6. Evolution of the SGS turbulence velocity contributions in a SN Ia simulation. The white contours represent the sections through the flame surface. The quantities presented in each of the four panels are: rate of production caused by the strain (left-top), specific Archimedian (buoyancy) force (top right), rate of dissipation (left bottom), and rate of diffusion (right bottom). (a) $t = 0.3$ sec, and (b) $t = 0.75$ sec. Reprinted from Ref. [173] with permission.

approach is to develop and solve transport equations for the higher order SGS moments [220–222]. This approach suffers from a lack of convergence with respect to the moment equations, similar to the classical closure problem in RANS [192]. Nevertheless, the larger number of coefficients to be calibrated, as well as the rich experience with engineering RANS modeling, can potentially offer more flexibility than Boussinesq type models. Some models combine RANS and moment methods for LES in a single multiscale approach that can relax from one methodology to another based on the level of SGS fluctuations [222,223]. These approaches, if successful, can account for nonuniform resolutions inherent in adaptive mesh simulations. However, for LES of reacting flows, the number of even the lowest order SGS moments involving the scalars (*i.e.* composition and thermodynamic variables) grows quickly, making the approach intractable. For this reason, the model has been used primarily for treatment of the SGS velocity.

Assumed SGS-PDF Methods: A remedy for moment methods when dealing with reaction kinetics involving many species is to augment them with PDF methods. These methods go back to the early work of Toor [224] and have been widely used in RANS [225]. They offer the advantage that all the single-point statistical information pertaining to the desired field is embedded within the PDF. The approach can be applied to the joint PDF of the composition variables or considering the full set of the primary variables. Therefore, once the PDF is known, the effects of composition fluctuations are easily determined. A systematic approach for determining the SGS-PDF is by means of solving the transport equation describing its evolution [226,227], which is rigorously derived from the governing equations. In this equation, the reaction source terms appear in a closed form. However, modeling is needed to account for transport of the PDF in the domain of the random variables. This transport describes the role of molecular/atomic effects on the evolution of the PDF. For supernova simulations, the form of the atomic transport terms is not fully known.

A major simplification is to assume the shape of the PDF *a priori*, usually in terms of the low order moments of the random variable(s), instead of solving a transport equation. Obviously, this assumption is *ad-hoc*, but it offers the possibility of performing calculations of practical flows, where accounting for full PDF transport is not feasible. In LES, this approach is pioneered by Madnia et al. [220,228] and has been used extensively by others; see *e.g.* Refs. [229–236]. The method has been used in conjunction with the “infinitely fast” and/or the equilibrium reaction models. In this case, the composition and thermodynamical variables are related to the “mixture fraction,” which is a conserved scalar variable constructed from the species’ mass fractions and the specific reaction model. This model is recommended when the structure of the flame can be assumed to be described by this mixture fraction, while the SGS velocity fluctuations are modeled separately. In this regard, the model is certainly an improvement over the no-model approach. However, it is virtually impossible to construct relevant multivariate assumed PDFs [229,237], limiting the use of the model for predictions of flows with non-equilibrium chemical reactions.

Filtered Density Function (FDF) Methods: Most (if not all) of the PDF models, as reviewed above, make the assumption of a β -density for the SGS-PDF of the mixture fraction [228,230–232]. This requires knowledge of the first two moments of the mixture fraction [220,233], which are usually determined by the moment method. An obvious drawback of the approach is the assumption that all the SGS composition and thermodynamic variables fluctuations can be described by a univariate PDF. The most systematic means of determining SGS-PDF is via the filtered density function (FDF) methodology [193,238]. The most sophisticated FDF closures available to-date are the pressure-energy-velocity-composition FDF (PEVC-FDF) [239], the frequency-velocity-scalar FDF (FVS-FDF) [240], and a simpler version (VS-FDF) which does not include the frequency [241,242]. Hydrodynamic closure in incompressible, non-reacting flows has been successfully achieved via the velocity-FDF (V-FDF) [243], and the closure which has been utilized by most other investigators only considers the scalar field (S-FDF and S-

FMD) [244]. This is the most elementary form of the model [184,245–249], and has experienced widespread applications for LES of a variety of reactive flows. Some examples are in Refs. [244,250–273]; see Refs. [198,274,275] for recent reviews. In almost all of these contributions, the FDF is considered for flows at low compressibility levels. As such, the effects of pressure fluctuations in the energy transport are negligible, and the latter is governed by a scalar equation similar to that for the composition. Some corrections to account for the effects of pressure in LES of strongly compressible flows have been attempted [239,276–278]. The FDF provides the most comprehensive form of SGS closures. In its simplest form, considering only the mixture fraction, the flamelet model is recovered. The degree of sophistication is increased by considering the joint FDF of more physical variables [275]. Along with that comes the complexity and higher computational cost [279]. This has been a deterrent for some to adopt the model for LES. However, with modern developments in Monte Carlo methods [257,264], FDF solvers have become very affordable. As such, the method would be the most comprehensive for SNe simulations.

The models surveyed below are primarily for SGS closures as pertaining to species and thermodynamic fluctuations, reaction rates, and/or the flame structure. Therefore, for these, additional modeling is required for the hydrodynamics.

Eddy Break Up (EBU) Models: These models are based on a simple idea originated in RANS [280,281]. In this approach, the inner structure of the flame is ignored and it is assumed that the reaction rate is directly determined by hydrodynamics [282]. In nonpremixed flames, the reactant conversion rate is related to the turbulence frequency. In premixed flames, the flame evolution is governed by the dynamics of a progress variable. This variable is typically the normalized temperature which goes from zero in the regions with fresh mixture and one in the fully burned fluids. Examples in combustion LES are given in Refs. [283–287]. Due to its simplicity, this approach can be readily employed for LES of SNe, and is expected to perform better than the no-model approach.

Laminar Flamelet Methods (LFM): When the reactions are fast compared to the turbulence time scales, the *flamelet* approach becomes appropriate. A laminar diffusion flamelet is defined as the region in the vicinity of the stoichiometric mixture fraction surface. This concept is due to Peters [288,289] and can be considered in either steady or unsteady contexts. In this model, all the composition and thermodynamic variables are parameterized in terms of the mixture fraction and other auxiliary variables. The joint statistics of the mixture fraction and these variables are specified in a very *ad hoc* manner [290]. In fact, in most cases, the specification is simply an assumption with very little justifications. However, due to its relative simplicity, the model has been used extensively in combustion research. For some examples, see Refs. [232,291–301]. The model can be effective for SNe simulations. In this way, the turbulent nuclear flame is described by the joint PDF of the mixture fraction and a reaction progress variable. With the specification of the joint PDF, or its determination by other means, some dynamical features of the flame can be predicted. These predictions are expected to be more accurate than those based on the no-model or the assumed univariate-PDF methods.

Conditional Moment Methods (CMM): Another methodology involving assumed PDFs is the conditional moment closure (CMC), originally due to Klimenko [302], and extended by Bilger [303,304]. This method considers the averages of the composition and thermodynamic variables, “conditioned” on the mixture fraction. The resulting equations feature the reaction source term conditioned on the mixture fraction. Typically, this term is approximated by using the mean conditional mass fractions and the conditional temperature. This is referred to as first-order CMC, as the higher order conditional moments are neglected. The CMC approach has shown some progress for LES [305–317]. This model is more sophisticated and complex than the flamelet model. Therefore, it is expected to provide a better approximation of the flame structure. For SNe simulations, this model can be

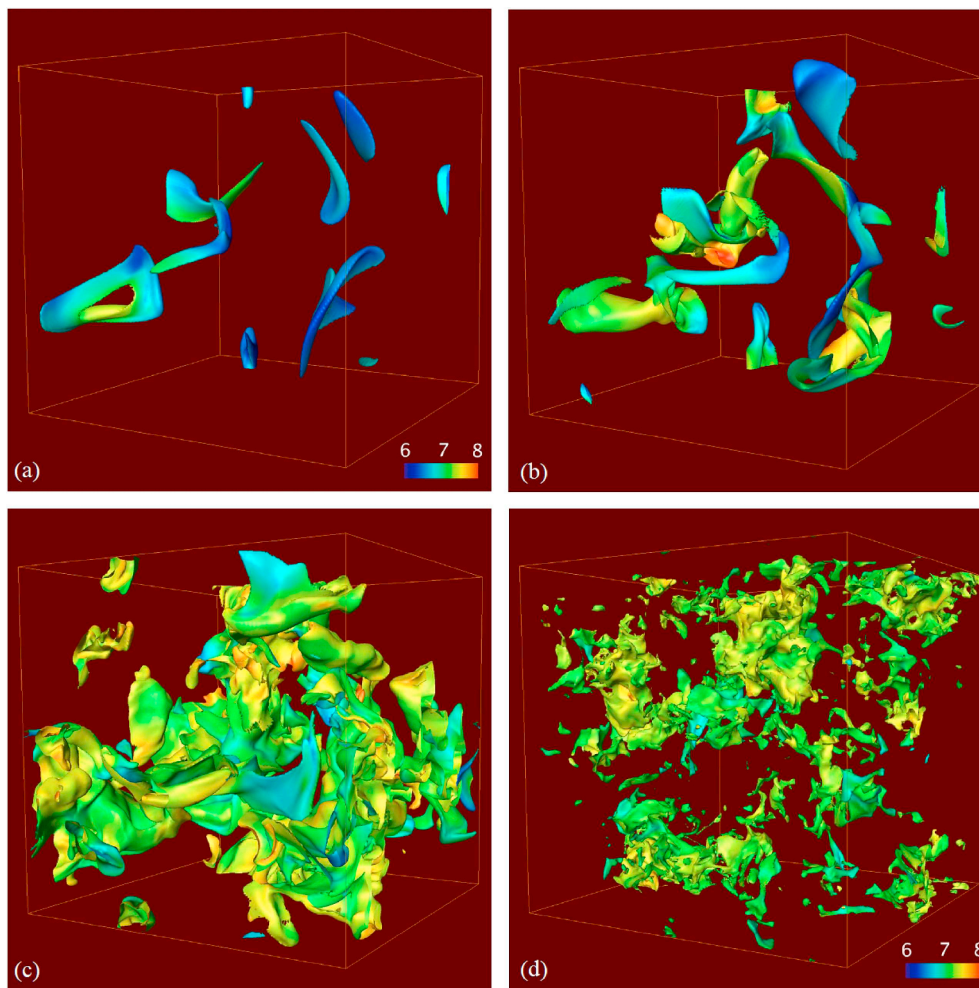


Fig. 7. LES of thermonuclear deflagration in a SN Ia in a cubic domain with periodic BCs. Shown are snapshots of the flame fronts with contour sections of the SGS turbulence velocity in logarithmic scaling. Parts (a) to (d) correspond to increasing times. Reprinted from Ref. [219] with permission.

more effective than the flamelet model if the spatial dimensionality of the conditional averages can be reduced.

Mapping Methods (MMC): A somewhat similar and newer approach is that of multiple mapping conditioning (MMC). This method was developed by Klimenko and Pope [318] and unifies the CMC and PDF approaches. In MMC, the species transport equations are conditioned on a subset of reference variables, N_m , where $1 < N_m < N_s$ and N_s is the number of species describing the reaction kinetics. This subset can include quantities such as the mixture fraction and other auxiliary variables. The composition is then represented by the joint PDF of the reference variables, and the conditional means of the remaining variables conditioned on the reference variables, for which a transport equation is constructed. The benefit is that it is not necessary to construct a joint PDF of dimension N_s which can be expensive. Also, a well chosen set of reference variables should yield better predictions than the traditional CMC. For recent applications of MMC, see Refs. [319–321]. This model is an improvement of the CMC and should be considered for SNe simulations.

One-Dimensional Models (LEM and ODT): The linear-eddy model (LEM) is a one-dimensional mixing and reacting closure originally due to Kerstein [322–328]. Implementation for LES involves two steps. The first is a turbulent stirring mechanism, modeled by a rearrangement of the scalar field (usually a progress variable) according to the *triplet map*. This represents the effect of turbulent mixing on the SGS due to a vortex of size ℓ , with $\kappa < \ell < \Delta_H$, where κ is the Kolmogorov length scale and Δ_H the LES filter size. The second step is the solution of a one-

dimensional unsteady diffusion-reaction equation. The LEM has been employed for the LES of a variety of reacting flows [329–331]. The extension of the method known as one-dimensional turbulence (ODT) [332–334] includes the velocity field as well [162,335–344]. Due to its simplicity, LEM has experienced usage in SNe Ia simulations. Refs. [199,200] have performed calculations with LEM and a 7 isotopes RN. Fig. 8 compares the carbon mass fraction and temperature profiles from a LEM simulation by Woosley et al. [199] and a 3D study by Aspdén

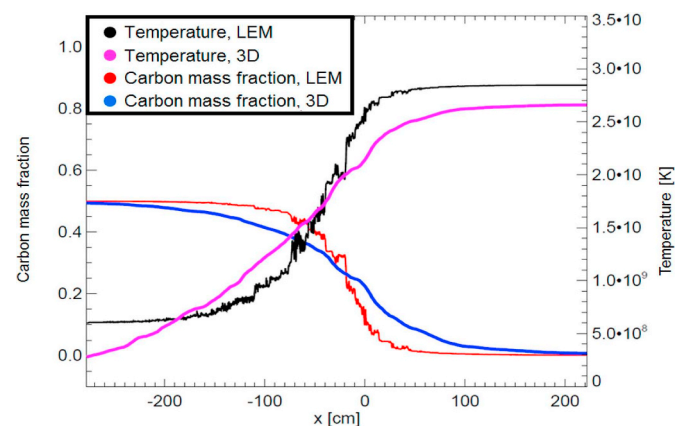


Fig. 8. Comparison between the flame results of a 1D LEM simulation [199] and a fully 3D simulation [160]. Reprinted from Ref. [199] with permission.

et al. [160]. The comparisons look promising, however, further studies are required to make a more comprehensive evaluation of LEM-ODT for SNe simulations, especially for more complex flame structures, as for example associated with instability development.

Interface and Level-Set Models: The original version of these closures is known as the G -equation model. In this “flame tracking” model, the flame thickness is considered negligible, and the flame front is associated with a constant property surface defined by $G = G_0$, where G is a signed distance function [195,345,346]. A transport equation for the propagation of this surface is solved. In this “geometric” representation, values of $G > G_0$ represent fully burned fluids, and $G < G_0$ represent unburned fluids [347]. This interface is tracked under the combined effects of hydrodynamics and reaction kinetics. The mathematical framework for the infinitely thin flame sheet dynamics is due to Shivashinsky [348]. Development of accurate schemes to determine the transport of a discontinuous flame sheet is at the heart of interface schemes, similar to shock fitting schemes in compressible flows. In reacting flows, the effects of exothermicity (thermal expansion and baroclinic torque) on the interface transport must be examined [349–351]. The numerical approach for tracking the interface was originally done through the volume of fluid (VOF) [352] method and its many subsequent variants. References [353–362] provide examples of LES via these models. For sample applications in SNe simulations, see Refs. [61,75,114,136,139,169–179,183]. As examples, Fig. 7 shows snapshots of the flame front for a thermonuclear SN simulation [219], and Fig. 9 shows deflagration to detonation transition with two different resolutions [179]. These models provide a simplistic representation of the SN burning topology. However, they cannot be trusted for reliable quantitative predictions.

Thickened Flame Methods (TFM): To extend the interface models to include more complex flame effects, it is a challenge to account for the flame structure, because the thickness of the flame is usually smaller than the LES grid size [195]. Some of the early attempts in this regards are developments of the so called “flame surface density,” “flame capturing,” and “flame wrinkling” models [363–366]. Some examples of using flame capturing models for SN Ia simulations are reported in Refs. [100,116,367]. In these models, the shape of the flame front (as depicted by the interface) is used to evaluate the filtered value of the reaction source terms. The most recent form of this geometrical modeling is TFM, originally suggested in Ref. [368]. In TFM, the idea is to consider a flame with the same laminar flame speed as the actual one, but have it artificially thickened in such a way that the combustion zone is resolvable by the computational grid. The method has shown some promise for a variety of combustion applications; see Refs. [369–378]. It has also been used in SNe Ia calculations with the MAESTRO code

[55]. The application of this model for LES of SNe is recommended and is expected to predict the flame structure better than the EBU and no-model approaches.

5. The role of DNS (Big) data

Direct numerical simulation (DNS), in which all the dynamically relevant scales of motion are fully resolved, without resort to SGS modeling or numerical stabilizing schemes, has been in the arena of turbulence research since the 1960's [379]. Since the pioneering work of Hill [380], the use of DNS has been growing at a very steady pace in the reacting flows research community. The rate of progress has been proportional to advancement, in computational capabilities, growing with each transition into a new era of computing; from scalar computing (1960–70's) to vector computing (80's) to parallel computing (90's) and to massively parallel computing currently at the petascale. Undoubtedly, this trend will continue with *exascale* computing on the horizon. With this evolution, the constitutive elements of DNS have been changing accordingly [381]. Today's standards and requirements for broad flow scale ranges limit DNS to mostly users with access to large supercomputers. These simulations are very expensive, but they do produce valuable data. Because of their size (typically of hundreds of terabytes), the task of data management has proven daunting and, in some cases, impossible. However, public databases, such as that hosted by Johns Hopkins University [382], offer access to multi-terabyte turbulence datasets of canonical turbulent flows, including analysis tools, for users worldwide. Recent additions facilitate the computation of statistics and extraction of features so that the flows can be analyzed remotely through the use of commodity hardware [383]. Such initiatives are costly, but they definitely offer tremendous scientific value for the broader scientific community. In particular, more efforts on *data sharing and citation framework* are needed to enable others to perform similar assessment of future closures.

One of the earliest DNS of reacting flows was conducted on 8^3 grids with transport of 2 chemical species [380]. Today, DNS is almost routinely conducted on $\mathcal{O}(10^8)$ - $\mathcal{O}(10^9)$ grid points with transport of $\mathcal{O}(10)$ species [384–387]. Some of the relatively more recent (post Y2K) DNS are those for premixed [385,388–400], non-premixed [401–405] and partially premixed [384,386,387,406–409] flames. For simple flows, current simulations are on par or exceed the Reynolds numbers of the corresponding physical laboratory experiments. These include homogeneous isotropic turbulence [410], Rayleigh-Taylor instability [411–413], and its homogeneous version [414], shear-driven turbulence [415], and even shock-turbulence interaction [416]. The range of scales that can be considered in these simulations are far from those in

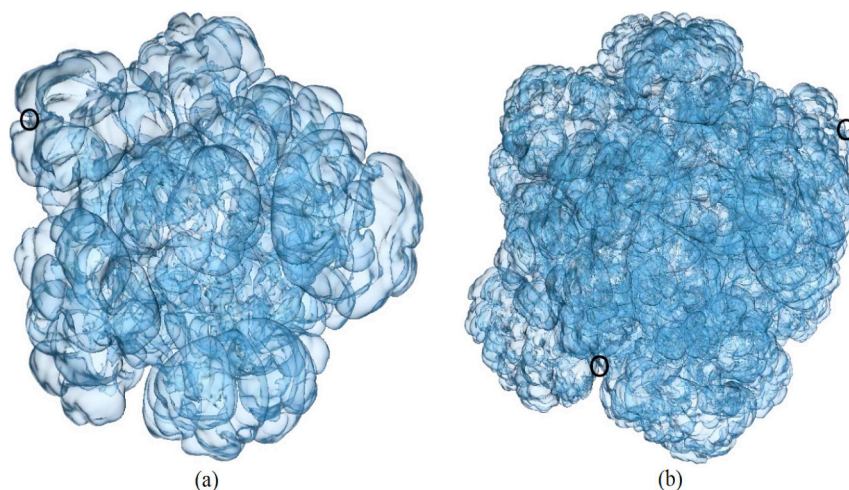


Fig. 9. Deflagration flame (transparent iso-surface) at the time of the first DDT in a SN Ia simulation. The DDT spots are encircled. a) Simulation with 256^3 grid cells. b) Simulation with 512^3 grid cells. Reprinted from Ref. [179] with permission.

astrophysical flows. As an example, a simple estimate [187] shows that the wall-clock time required for DNS of solar convection over one dynamical time scale would be larger than the current age of Universe on today's fastest supercomputer! However, for basic canonical turbulent flows, it is believed that there is an asymptotic regime such that further increase of the Reynolds number (i.e. the range of scales) does not yield significant changes of the turbulence characteristics [417–419]. As the flow develops an inertial range, the energy cascade becomes localized [420] and not influenced by the viscosity. From this point of view, it is tempting to assume that numerical dissipation in Euler simulations can act in a similar way and allow the development of an inertial range, so that the numerical solution is close to the physical solution when the grid is fine enough. However, developing a power law range in the spectrum is not proof of the emergence of an inertial range [421] and proving cascade locality in the presence of numerical viscosity/diffusion may be impossible in general. Many of the current high resolution DNS are well beyond this “mixing transition.” Evidence for the convergence of the flow characteristics with decreasing the values of the transport properties has also been found in flows as diverse as magnetic reconnection [422] or Richtmyer-Meshkov instability [423]. In other instances, the concept of separation of scales can also be used to justify the relevance of finite scale range simulations to practical applications. For example, when the shock wave thickness is much smaller than the flow scales, the results become independent of the shock profile. In this case, even though the Navier-Stokes description breaks around strong shocks, it can still accurately predict the shock-turbulence interaction [416]. Simulations using shock-capturing techniques also become accurate when the numerical shock thickness is much smaller than the flow scales [424]. This concept has also been used in reacting simulations related to SNe Ia applications by resolving the structure of the flame, without fully resolving the turbulence [160,425]. Therefore, current high resolution DNS can be used to test and develop turbulence models targeting practical applications with much larger Reynolds numbers.

In turbulent combustion, DNS data have been most useful to: (1) assess the performance of turbulence sub-closures, (2) appraise the overall predictive capability of turbulence models, and (3) enhance our fundamental understanding of the flow physics. For appraisal of closures, DNS data can be, and have been, used for both *a priori* and *a posteriori* assessments. For the former, the performance of the sub-elements of the SGS closures are tested against DNS data. For the latter, the final predictions are assessed by comparisons with (statistically averaged) DNS data. In idealized/simplified cases, it is possible to also use conventional laboratory data for the latter, but not for the former.

Traditionally, the construction of turbulence models for complex flows has been performed using a hierarchy of flows starting from homogeneous isotropic turbulence, separately addressing features such as turbulence production mechanisms, then verifying the closures in flows with coupled physics and through comparisons with laboratory data [426]. Direct simulations also have the advantage of being able to design flows to target specific physical effects, that are otherwise difficult or even impossible to isolate in physical experiments.

In all validations, the flow & thermonuclear reaction parameters should be identical in LES and DNS, but the grid resolution in DNS would obviously be higher. For demonstration, Fig. 10 shows the comparison of the flame surface in a shear flow configuration as obtained by DNS (Fig. 10(a)) and LES-PDF (Fig. 10(b)) [427]. It is shown that the DNS results are more rugged and include more of the fluctuations at the small scales; whereas LES captures the overall large scale structures. Another recent example of *a priori* DNS assessment is shown for a reacting shear flow in Fig. 11(a). This figure shows the topology of interaction of the flame (denoted by red iso-surfaces) with vortical structures (white) and the turbulent/non-turbulent interfaces (orange). These data are statistically sampled, with an example shown in Fig. 11(b). Here, scatter plots are presented of the SGS kinetic energy vs. the viscous dissipation. Due to the large size of the simulation, approaching the “mixing transition,” the results suggest an improved scaling of the viscous dissipation compared to the scaling previously used in the turbulence literature. An assessment of this type via laboratory experiments is very difficult.

6. Concluding remarks

Accurate modeling of the small scale dynamics of thermonuclear flames is essential to understand the physics of SNe Ia. There has been significant progress in developments of computational methods and physical models for SN simulations. These advances have been encouraging; however, the state of the art is far from satisfactory. It is widely understood that LES will constitute the primary means of computational modeling of SN flames. Nevertheless, the operation procedure of LES for SNe (or any other astrophysics application) is markedly different from those in engineering, atmospheric, oceanic, and other fields. Unlike these fields, it is not feasible to conduct DNS for actual model validations under SN conditions. The same goes for empirical data; as most of such measurements cannot probe the SN parameter ranges and matter conditions. Thus, there is a significant jump from canonical flows to SN conditions that needs to rely on hypotheses such as “mixing transition” or convergence with respect to the

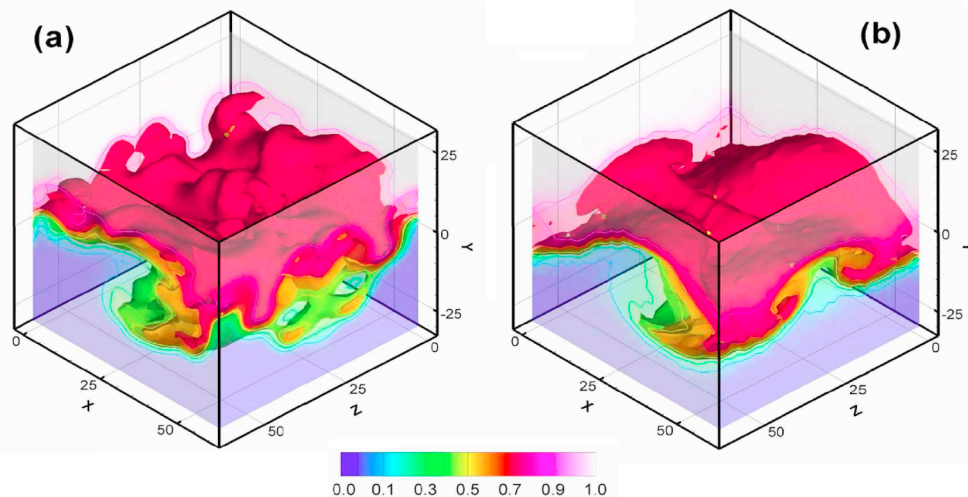


Fig. 10. Isosurfaces of the flame sheet in a shear flow: (a) DNS and (b) LES-PDF. Reprinted from Ref. [427] with permission.

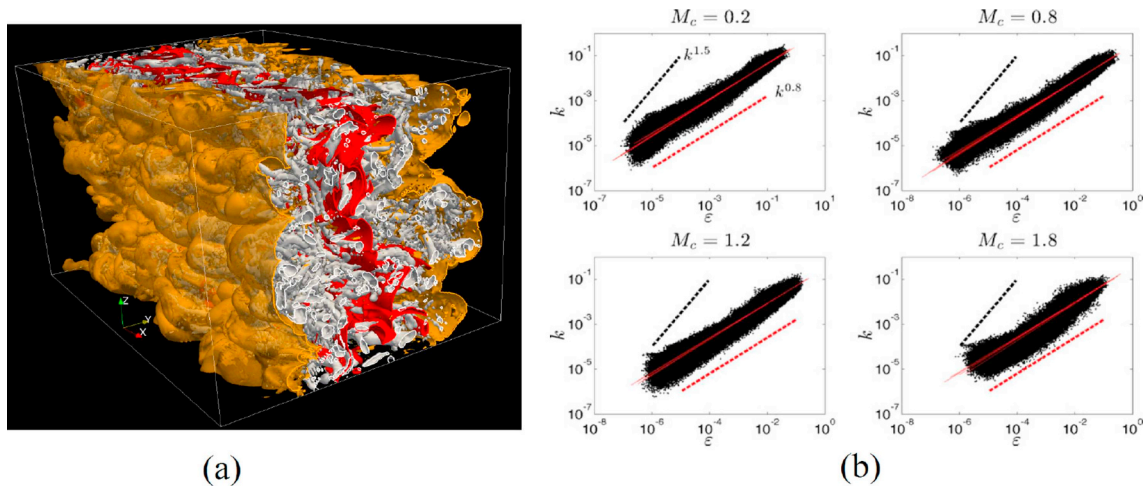


Fig. 11. DNS a compressible reacting mixing layer [428]: (a) 3D view and streamwise cut: turbulent/non-turbulent interface (orange translucent surfaces), flame (red), and regions of high vorticity magnitudes (gray). (b) Scatter plot of SGS viscous dissipation and SGS kinetic energy (k). Solid blue lines denote the power-law fitting of the scatter data for several times during the self-similar state. Broken black and blue lines denote $\propto k^{-1.5}$ and $\propto k^{-0.8}$, respectively [234]. Reprinted from Refs. [234,428] with permission.

Reynolds/Schmidt numbers for relevance. In the continuing efforts toward developments of more accurate models, several challenges need to be overcome. Some of these are discussed in recent reviews; e.g. Refs. [22,50,82,144]. Here, we add a bit more to these discussions (in no specific order):

- Specification of initial conditions remains a serious challenge, since the progenitor structure and the ignition process cannot be directly observed and should be modeled. A good characterization of SNe Ia progenitor systems is yet to be established. The astronomical identification of a progenitor can help with suggesting potential scenarios but it cannot resolve all of the initialization problems. Extensive simulations using higher accuracy models may help discriminate among possible known scenarios and also explore new/alternate possibilities. Moreover, recent advancements in uncertainty quantification (UQ) and sensitivity analysis may help in defining some confidence intervals on the estimate of the quantities of interest, e.g. Ref. [75] evaluates the impact of deflagration strength on ^{56}Ni production.
- Accurate description of the radiation processes, responsible for spectrum formation in the evolving ejecta, is still lacking. In some cases, the calculations can be performed as a post-processing step on the ejecta that have been dynamically simulated into the homologous phase. However, for scenarios where the ejecta interact with a dense environment, the ongoing dynamics need to be considered. For either case, the radiative transfer calculations become complicated by several factors: the ejecta material is rich in heavy elements, hence a large number of bound-bound transitions needs to be taken into account, the large expansion velocities have to be considered, and non-LTE effects become increasingly more significant as the ejecta expand. Consequently, modeling of spectra, especially at the latest observable epochs (200–300 days post explosion) critically depends on the microphysics and quality/quantity of the atomic data (radiative and collisional) available for the necessary ions.
- Proper representation of instabilities for propagation of deflagrations is still an open issue. Hydrodynamic instabilities, and in particular the Rayleigh-Taylor instability, represent the main early time acceleration mechanism for the flame, and significantly affect mixing and burning rates. This eventually alters some intrinsic properties of the ejecta like the kinetic energy and the nature and distribution of the synthesized isotopes. The problem is compounded by the fact that even the growth rate of classical Rayleigh-

Taylor instability is still under debate in the presence of contamination by long wavelengths in the initial conditions [413].

- Typically, magnetic fields are not explicitly considered in SNe Ia calculations. This is primarily due to computational limitations. However, small scale physics such as the Hall and Biermann battery effects can change the electron contribution to heat conduction [422]. In addition, recent simulations suggest that magnetic fields may have an inhibitory role on the Rayleigh-Taylor instability in the early stages of the flame formation, reducing the mixing and, thus, altering the explosion dynamics and element abundance [84]. High magnetic fields have been suggested as a solution for the discrepancy between observations and supernova models using a flame speed based on the fully developed Rayleigh-Taylor instability [84]. This is an intriguing possibility that requires further studies.
- A more accurate approach for SGS modeling as pertaining to thermonuclear reactions coupled with strong turbulence is needed. Most of the models in current use are somewhat primitive and do not provide a reliable representation of turbulence-reaction kinetics interactions. The ultimate means of such modeling via FDF methods may not be feasible computationally within the foreseeable future. However, simpler models such as TFM or LEM-ODT could be useful and are recommended. In this regard, representative small scale DNS can be used to construct the flamelet library for some realistic cases.
- Implementation of higher order computational schemes is desirable. The majority of current SN simulations are by means of low order numerical discretization schemes. There has been significant recent progress in the development spectral discretizations, including spectral-element and discontinuous Galerkin (DG) methods [429–432] for high-speed, compressible flows. These methods are now capable of dealing with sharp gradients, and have the potentials to be successful if coupled with an appropriate SGS model.
- The use of machine/deep learning techniques [433–435], including data-driven and reduced order modeling [436–438] are recommended, especially for RNs. When applying these techniques to large datasets, there is an opportunity to obtain new insight into the physics or capture hidden trends [439,440]. As an example, an artificial neural network with the ability to tabulate nonlinear function using minimal memory usage has been used in terrestrial combustion for modeling of the reaction rates [441]. In reduced order modeling, the goal is to reproduce specific characteristics of a system with a highly reduced computational cost. Although most of the reduced order methods are system dependent, some are capable

of capturing rare events [442]. This can be quite interesting in SN simulations, where an important challenge is prediction of abrupt transitions occurring on length and time scales far below CFD-resolved scales.

Acknowledgment

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